JACK NAIMER

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EDUCATION & TRAINING

Engineering Science – Machine Intelligence Major, University of Toronto 2021-2025

- Rigorous program famous for its intense academic challenge.
- Wide range of machine learning specialty courses as well as advanced courses in mathematics, physics, biology, and chemistry.
- Dean's Honours List – Winter 2024, Fall 2023, Fall 2021

Math and Computer Science (MaCS) enriched high school program, William Lyon Mackenzie Collegiate Institute 2017-2021

- Achieved 5/5 on the college-level AP Computer Science exam in 10th grade.

PUBLICATIONS

Preprint (arXiv:2410.15185): "Semantically Safe Robot Manipulation: From Semantic Scene Understanding to Motion Safeguards"– In review, *IEEE Robotics and Automation Letters*.

WORK EXPERIENCE

Undergraduate Research Student, Dr. Bo Wang, University of Toronto/UHN March 2024-Present (part-time)

Dr. Bo Wang is Chief AI Scientist of the University Health Network (UHN), Canada's largest research hospital, and Assistant Prof. of Comp Sci at University of Toronto.

Writing undergraduate thesis developing methods for generative AI genomic foundation model interpretability. Uncovering AI and genomic insights.

Co-leading development and deployment of AI system in hospital to predict surgical durations and improve operating room scheduling which reduces # of cancelled surgeries and improves care. Planning prospective clinical evaluation.
First author of manuscript (in progress) for *JAMA Cardiology*.

Undergraduate Research Student, Dr. Michael Hardisty, Sunnybrook Sept 2024-Present (part-time)

Implementing timeseries models to predict musculoskeletal degeneration in prostate cancer patients. Enables personalized treatment. Co-authoring manuscript.

Undergraduate Researcher, Dr. Angela Schoellig Learning Systems and Robotics Lab (LSY Lab), Technical University of Munich, Germany. May-August 2024

Preprint (arXiv:2410.15185)- Developed AI-based robotic semantic safety filter. Used LLMs and VLMs to detect safety issues in a scene and generate robotic constraints.

Developed a Lego-building robot using Franka robot arm. Focused on AI-vision components including brick detection and pose tracking using foundation models.

Data Science Intern, Marsh, New York office. June-August 2023

Developed a machine learning model for risk analysis in the healthcare field, the first of its kind in the field. Enables hundreds of hospitals to accurately assess their risk.

Research Student, SickKids Hospital with Dr. Sasha Litwin. May-June 2023

Developed an AI chatbot designed to answer a parent's fever questions, improve at-home treatment, and prevent unnecessary ER visits.

Research Student, Dr. Mike Fralick's Lab, Mount Sinai Hospital. May 2022-May 2023

Developed AI tools for healthcare. Full-time during summer, rehired to continue part-time during school year. Projects included:

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ChatBotMD

Created BERT-based AI chat bot prototype for clinical use, developed custom decision-making algorithms. Created chat bot website from scratch, set up networking, and run on AWS server with NGINX. Sole programmer.

Medical Literature Tool

Created website that uses my NLP and LLM approach to inform doctors of advances in medicine. Finds relevant papers and retrieves key information.

jrnowl.com

Using ML to automatically find journal publishing guidelines from medical journal websites for use on jrnowl.com

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

Academic Director & Academic Paper Read Group Mentor, University of Toronto Machine Intelligence Student Team (UTMIST). 2022-2023

Led monthly machine learning lectures. Topics taught included CNN, ConvNext, GNN, Variational Autoencoder, Attention, Transformer, etc.

Software Team Member, University of Toronto Design League Robodog Team. 2022
Responsible for inverse kinematics, and gait programming using ROS.

Competition Team Member, University of Toronto Concrete Toboggan Team 2021-2022

Satirical Newspaper Co-President, William Lyon Mackenzie CI. 2020-2021

Math and Computer Science Program Peer Mentor & Calculus Tutor 2019-2021

Camp Counsellor, Green Acres Day Camp. Led activities for campers age 5-8 Summer 2019

TECHNICAL SKILLS

- Python, Julia, C, C++, Java, Matlab, Linux, ROS, Docker, HTML, CSS, SolidWorks
- LLMs, NLP, Computer Vision, PyTorch, JAX, Tensorflow
- University of Toronto Machine Shop Certification (2021)

AWARDS & ACHIEVEMENTS

Engineering Science Global Research Opportunities Program, Highly competitive scholarship among Engineering Science students. For summer research in Angela Schoellig's LSY Lab in Munich. 2024

BioTalent Health and Biosciences Student Work Placement Wage Subsidy, For summer research in Dr. Mike Fralick's Lab. 2022

Admission Scholarship, Faculty of Applied Science and Engineering, University of Toronto. $2000 2021

Advanced Placement (AP), 5/5 Computer Science A (2019), 5/5 Calculus AB(2021)

DECA Provincial Finalist & Case Award, Business Finance Series 2021

Honourable Mention, Queen's Model United Nations Conference 2021

3rd Place, LyonHacks Hackathon, William Lyon Mackenzie Collegiate Institute 2021

DECA Regional Top 60, Business Finance Series 2020

References

Dr. Bo Wang - UHN Chief AI Scientist. University of Toronto Assistant Professor
bowang@vectorinstitute.ai

Dr. Angela Schoellig – Robotics Institute Germany Coordinator. Alexander von Humboldt Professor for Robotics and AI at TUM angela.schoellig@tum.de

Dr. Sasha Litwin – SickKids Emergency Medicine Physician, University of Toronto Assistant Professor sasha.litwin@sickkids.ca